CS3631 – Project

Last updated - 30th June 2026

1. Project Overview

This is a group-based research project, with each t**eam consisting of 5 students**.

The objective of this project is to identify, design, implement, and evaluate a novel or improved deep neural network framework for a real-world problem and to prepare a research paper suitable for submission to a local or international conference.

Students will experience the complete lifecycle of deep learning research, including:

- Problem formulation
- Dataset acquisition and characterization
- Baseline model selection and implementation
- Deep neural network design and experimentation
- Model evaluation and ablation studies
- Computational efficiency analysis
- Research writing and academic dissemination

This project simulates a real-world applied deep learning research project.

2. Dataset Requirements

Students may use one of the following approaches.

2.1 Extension of CS3121 Datasets (Preferred)

Students are strongly encouraged to build upon datasets collected during the CS3121 – Introduction to Data Science project by:

- Extending the temporal coverage of the dataset
- Incorporating additional data modalities
- Collecting additional samples
- Integrating complementary datasets
- Reformulating the problem into a more challenging task

2.2 Existing Public Datasets

Students may also use datasets from platforms such as Kaggle, UCI, HuggingFace or Government websites or use research benchmark datasets or public APIs.

3. Project Objectives

Students are expected to pursue one of the following research objectives.

1. Design and Develop a new DNN framework for a new dataset
2. Design and Develop a new DNN framework/ modify an existing one to improve predictive performance for an existing dataset
3. Optimize the DNN by reducing training time, inference latency, memory consumption and/or energy consumption.

4. Baseline Model Requirement (Mandatory)

The **first task** of every project is to identify and implement the most obvious baseline deep learning architecture for the selected dataset.

Data Type	**Baseline Model**
Images	CNN
Sequential data	LSTM/GRU
Time series	LSTM/Transformer
Text	Transformer
Graphs	GCN/GAT
Audio	CNN/LSTM
Multimodal	Late-fusion baseline

The baseline model serves as the reference point for evaluating the proposed contributions.

5. Model Constraints

The goal of this project is to design/develop novel and/or efficient deep learning frameworks, rather than simply applying large pre-trained models or simply adapting existing models. Therefore, the following approaches **are not permitted**.

- Agentic AI systems
- Multi-agent systems
- LLM-based approaches
- Prompt engineering approaches
- End-to-end use of commercial LLM APIs
- Large foundation model fine-tuning requiring specialized hardware

Instead, you should use

- MLPs
- CNNs
- RNNs and variants such as LSTMs, GRUs
- Transformers (Preferred over RNNs and LSTMs)
- GNNs
- Autoencoders
- Contrastive learning
- Self-supervised learning
- Multimodal learning

6. Computational Constraints

All experiments must be executable using free computational resources such as Google Colab and Kaggle or Personal resources (if available).

Students should not assume access to:

- Multi-GPU clusters
- HPC infrastructure
- Large cloud computing budgets

Projects should be designed to train within reasonable resource limits.

7. Use of Large Language Models

Large Language Models may only be used as a preprocessing tool.

Permitted examples:

- Generate sentence embeddings
- Generate document embeddings
- Generate code embeddings
- Generate feature representations

However:

- The generated embeddings must be stored and saved prior to training.
- The LLM itself must not be part of the trainable end-to-end architecture.

8. Expected Research Contributions

Possible contributions include:

- Architecture
    - Novel DNN architecture
    - Novel attention mechanism
    - Novel fusion strategy
- Learning
    - Novel loss function
    - Novel training strategy
    - Novel regularization technique
- Representation
    - Novel feature representation
    - Novel graph representation
- Efficiency
    - Model compression
    - Knowledge distillation

    - Efficient inference

9. Research Paper Requirement

Each group must prepare a conference-style research paper containing:

- Abstract
- Introduction
- Related Work
- Proposed Framework
    - Implementation Details
- Experimental Setup
    - Dataset Description (An overview only.)
    - Baseline Models
    - Experiments
        - Ablation Studies
        - Comparative Analysis with the existing methods
        - Hyper-parameter Tuning experiments
        - Computational Analysis
        - Other experiments to demonstrate the advantages of the proposed approach
- Discussion
- Conclusion
- References

10. Marks Allocation Based on Publication Outcome

Paper Accepted

- Accepted at a high-rank / high h-index conference: 100%
- Accepted at any recognised conference: 95%

Paper Rejected

- The paper will be reviewed by module lecturers (**Phase 1**, **Phase 2**, and **Phase 3**)
- Marks awarded using the research paper rubric. However, the maximum you can obtain is 84%.

Paper Never Submitted

- The paper will be reviewed by module lecturers (**Phase 1** and **Phase 2**)
- Marks awarded using the research paper rubric. However, the maximum you can obtain is 50%.

11. Group Work & Individual Responsibility

Each group must consist of **exactly 5 students**. Although this is a group project, **each student must make a significant and identifiable contribution.** Each student is expected to contribute meaningfully to at least one stage of the DNN framework development process (e.g., feature engineering, DNN architecture design, implementation, experiments). Mere writing, formatting, or minor edits alone are not considered sufficient contributions.

Each team must submit a colour-highlighted version of the research paper for grading purposes, in addition to a clean version for conference submission. Assign one distinct colour to each team member (five or six colours depending on group size) and use these colours consistently throughout the document to indicate individual contributions. Students should highlight the sections to which they made substantial contributions, including text they wrote, figures or visualisations they created, tables they generated, methodology they implemented, and experimental analysis they conducted.

Since research writing and technical implementation may be carried out by different team members, short margin comments should be added where necessary to clarify roles. For example, a student who implemented a machine learning model may be different from the student who wrote the corresponding section. In such cases, clearly indicate both contributions. You may also add short comments to clarify specific technical contributions and joint contributions (if multiple members worked together).

12. Deliverables and Timeline (Submit to relevant Moodle links)

- **Phase 1: Project Proposal - Week 5 (11.59 PM August 2, 2026) - 25% of Project Marks**
    - The purpose of the proposal is to assess feasibility, scope, and suitability, and to prevent overambitious or underdefined projects.
    - The proposal must include:
        1. **Problem Motivation**
            - Clearly define the real-world problem (e.g., High accuracy requirement, limited resource availability etc)
            - Explain why it matters and who benefits
        2. **Identified Data set/Data sets**
            - An overview of the dataset
        3. **Baseline Model**
            - Model Architecture
            - Implementation details
            - Baseline Model Experimental Results
                i. Including hyper-paramter tuning
        4. **Proposed Contribution**
            - Expected novelty
            - Expected improvement
        5. **Experimental Plan**
            - Evaluation metrics
            - Baselines
            - Resource requirements
            -
    - **Page limit** - 2 Pages excluding references.
    - **Template** - ACM Conference Proceedings Primary Article Template - https://www.overleaf.com/latex/templates/acm-conference-proceedings-primary-article-template/wbvnghjbzwpc

- **Phase 2: Short Paper - Week 8 (11.59 PM August 23, 2026) - 25% of Project Marks**
    - Short paper submission
        - **Proposed architecture**
        - **Initial evaluation of the proposed architecture**
        - **Comparison with the baseline model**
    - **Page limit** - 4 Pages excluding references.
    - **Template -** ACM Conference Proceedings Primary Article Template - https://www.overleaf.com/latex/templates/acm-conference-proceedings-primary-article-template/wbvnghjbzwpc
- **Phase 3: Complete Research Paper, LaTeX Zipfile and Code - Week 12 - (11.59 PM April 26, 2026) - 34% of Project Marks**
    - **Authorship requirements**
        - First five (or 6) authors - The group members. You can order as you prefer. Ideally, we order by contribution to the paper: the student who contributed the most should come first.
        - The next co-authors should be your assigned TAs.
        - Last two co-authors - Prof Dulani and Dr Sandareka. Please note that we do not include titles in the author list.
    - **Page limit** - Depends on the conference you chose.
    - **Template -** Depends on the conference you chose.
- **Conference notification/ conference submission proof - 11.59 PM November 15, 2026**

Important

- The project should demonstrate research thinking, not merely model application.
- Novelty may arise from:
    - the architecture,
    - the learning strategy,
    - the data representation,
    - the optimization strategy,
    - the explainability framework,
    - or the computational efficiency improvements.
- Students are encouraged to start from a strong baseline and justify every design decision through experimentation.
- All work must be the original contribution of the students.